

# Task 1 – AI Problem Design Document

Project: EduIntent AI – Intelligent Student Query Classification System

| Parameter           | Specification / Result                                                                            |
|---------------------|---------------------------------------------------------------------------------------------------|
| Project Name        | EduIntent AI – Student Query Intent Classifier                                                    |
| AI Task             | Multi-Class Text Intent Classification                                                            |
| Unique Features     | Explainable AI (XAI), Ambiguity Detection, Low Confidence Guardrails, Code-Mixed Language Support |
| Academic Categories | 9 Core Computer Science & Administrative Subjects                                                 |
| Dataset Size        | 450 Hand-curated Questions (50 per category)                                                      |
| Best Algorithm      | Calibrated Linear SVM (TF-IDF Unigrams + Bigrams)                                                 |
| Test Accuracy       | 87.78% (Target: ≥ 85%)                                                                            |
| Macro F1-Score      | 0.8746 (Target: ≥ 0.80)                                                                           |
| 5-Fold CV Accuracy  | 89.72% (± 0.02)                                                                                   |

## 1. Problem Statement & Unique System Design

Students frequently ask academic queries across subjects (Python, ML, Deep Learning, DBMS, OS, Networks, Flutter, DAA, General Academic Queries). Manually categorizing queries is inefficient. EduIntent AI automatically categorizes queries, displays confidence scores, extracts key contributing terms (XAI), detects ambiguous cross-domain queries, and alerts users on low-confidence inputs.

**Gujarati Context:**  
Students █████ academic queries █████ ███. EduIntent AI model query █████ automatically █████ category, confidence score, █████ key terms █████.

## 2. Benchmark Comparison Across Algorithms

| Model Algorithm             | Test Accuracy | Macro F1 | 5-Fold CV |
|-----------------------------|---------------|----------|-----------|
| Baseline (Majority Class)   | 11.11%        | 0.0222   | 11.11%    |
| Naive Bayes (MultinomialNB) | 85.56%        | 0.8547   | 87.50%    |
| Logistic Regression         | 83.33%        | 0.8275   | 86.39%    |
| Linear SVM                  | 87.78%        | 0.8746   | 89.72%    |

## 3. Qualitative Error Analysis Table

| Question                                    | Expected      | Predicted        | Reason / Analysis                                      |
|---------------------------------------------|---------------|------------------|--------------------------------------------------------|
| 'Neural network optimization loss function' | Deep Learning | Machine Learning | Shared ML/DL vocabulary (loss functions, optimization) |

|                                             |                 |                  |                                                     |
|---------------------------------------------|-----------------|------------------|-----------------------------------------------------|
| 'B+ Tree structure in file indexing'        | DBMS            | DAA / Algorithms | Shared data structures & indexing terminology       |
| 'How to plot correlation matrix in Pandas?' | Python/Data Sci | Python/Data Sci  | <div>■ Correct (High precision keyword match)</div> |

Figure 1: Benchmark Comparison Chart

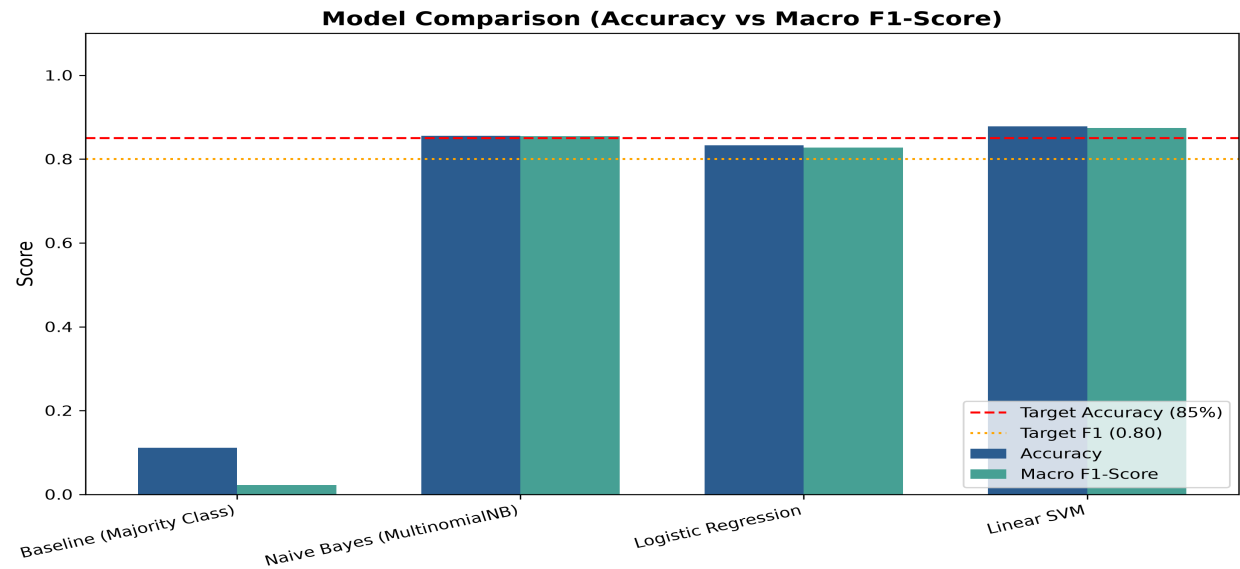


Figure 2: Confusion Matrix (Linear SVM)

